Priyanshu Singh

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github.com/priyanshu-48

Education

B.Tech in Electronics and Communication Engineering **2022 – 2026**
Vellore Institute of Technology, Vellore *CGPA: 8.39/10*

AISSCE (Class XII) **2022**
Delhi Public School, Siliguri *85.2%*

AISSE (Class X) **2020**
Delhi Public School, Siliguri *95.4%*

Skills

Languages: TypeScript, JavaScript, Python, C++, SQL, Bash
Frameworks & Libraries: Node.js, Express.js, Fastify, React.js, Django, Tailwind CSS, BullMQ, Drizzle ORM
Databases: PostgreSQL, Redis, MongoDB
Tools, Cloud & DevOps: Docker, Docker Compose, Git, GitHub, GitHub Actions (CI/CD), Render, Vercel, Postman, Vitest, Testcontainers, k6, Claude Code, Chrome Extensions (Manifest V3)
Concepts: Data Structures & Algorithms, Object-Oriented Programming, RESTful APIs, WebSockets, JWT Authentication, Event-Driven Architecture, Message Queues, Unit & Integration Testing, Cloud Deployment

Projects

Real-Time Notification Service GitHub

- Architected a multi-tenant notification service with **22 documented API operations** and a **WebSocket** stream, delivering email and real-time in-app notifications through a **BullMQ** job queue.
- Engineered crash-safe, at-least-once delivery using **idempotency keys** and **exponential-backoff retries**; a chaos test killing a worker mid-send (20 notifications in flight) showed **0 lost, 0 duplicated**.
- Load-tested with **k6** at **18,000 notifications/min, 0 loss, 24 ms p95** (local run, mock email) and wrote **86 tests (79.8% coverage; 8 suites on real PostgreSQL/Redis via Testcontainers)** run in **GitHub Actions CI/CD**.
- **Tech Stack:** TypeScript, Node.js, Fastify, PostgreSQL, Redis, BullMQ, WebSockets, React, Docker, Render, Vitest, k6.

Distraction Tracker GitHub | Video Demo

- Built a full-stack browsing-analytics platform with a **Chrome Extension (Manifest V3), React, Express.js**, and **PostgreSQL**; batched, UUID-deduplicated uploads make retries safe.
- Optimized the original analytics queries from **1,176 ms to 5.7 ms (206x) on 1M rows** by adding a composite index and rewriting date filters as range queries.
- Hardened auth with **JWT** in httpOnly cookies, **bcrypt**, token revocation, and CSRF protection; **986 automated tests** (including a 16-route authorization table) run in a 4-job **GitHub Actions** pipeline.
- **Tech Stack:** React, TypeScript, Tailwind CSS, Node.js, Express.js, PostgreSQL, Manifest V3, JWT Auth, Docker.

AI Course Generator GitHub | Live Demo

- Developed a course generator on the **Gemini and YouTube APIs** and cut generation latency **50% (p50 7.2 s to 3.6 s, simulated APIs)** by resolving 28 YouTube lookups concurrently (**5.3x faster** on real YouTube calls).
- Added **Redis** caching (a warm cache removed all 28 YouTube calls per repeat course), retries with backoff (success under 20% injected failures **81% to 99.5%**), rate limiting, and Gemini model fallback for free-tier quota limits.
- Wrote **63 backend tests (94% coverage)** and 19 frontend tests with **GitHub Actions CI**; cut the **Docker** image **59% (199 to 81 MB)** and halved build time.
- **Tech Stack:** Django REST API, React, MongoDB, Redis, Tailwind CSS, JWT Auth, Docker, Gunicorn, pytest, Vitest.

Certifications & Achievements

- Earned the **LeetCode Knight** badge (contest rating **1915**, Top 4.26%). [Link]
- Ranked **193** in LeetCode Biweekly Contest 171 and **465** in Biweekly Contest 167.
- Solved **650+ DSA problems** across LeetCode, GeeksforGeeks, and other platforms. [Link]
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate. [Link]